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Declarations under Rule 4.17:
— *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*
— *as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))*

(54) **Title:** SOLID STATE FORMS OF DELGOCITINIB AND PROCESS THEREOF

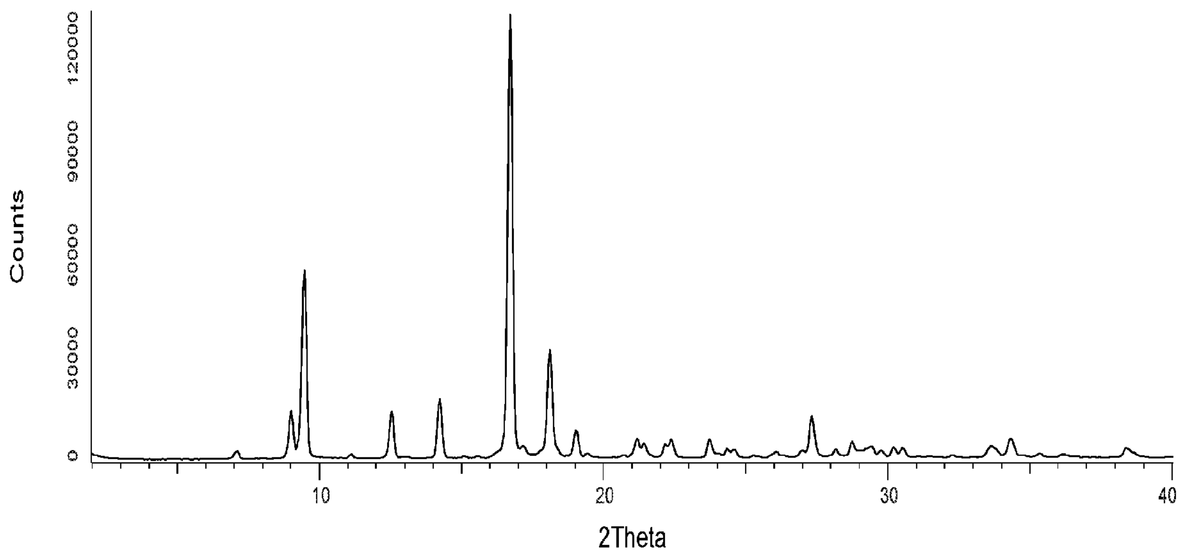

Figure 1. An XRPD pattern for crystalline Delgocitinib Form DL1.

(57) **Abstract:** The present disclosure relates to Delgocitinib solid state forms, processes for preparation thereof, pharmaceutical compositions thereof, and methods of use thereof.

[Continued on next page]

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Published:

— *with international search report (Art. 21(3))*
— *before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))*
— *in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE*

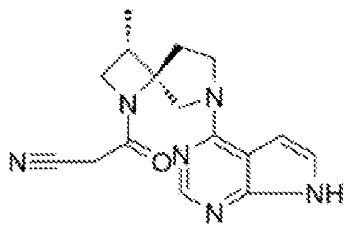
SOLID STATE FORMS OF DELGOCITINIB AND PROCESS THEREOF

FIELD OF THE INVENTION

[0001] The present disclosure relates to Delgocitinib solid state forms, in embodiments crystalline polymorphs or salts of Delgocitinib, processes for preparation thereof, pharmaceutical compositions thereof, and methods of use thereof.

BACKGROUND

[0002] Delgocitinib (JTE-052) which has the chemical name 3-[(3*S*,4*R*)-3-methyl-7-(7*H*-pyrrolo[2,3-d]pyrimidin-4-yl)-1,7-diazaspiro[3.4]octan-1-yl]-3-oxopropanenitrile, is a pan-Janus kinase (JAK) inhibitor and is approved for treatment of atopic dermatitis (AD) in Japan. As described in U.S. Patent No. 8,609,647, Delgocitinib has the following chemical structure:



[0003] Delgocitinib preparation is disclosed in U.S. Patent Nos. 10,822,354 and 8,609,647 and Delgocitinib polymorphs are described in U.S. Patent Nos. 11,339,181, and 11,312,728.

[0004] Polymorphism, the occurrence of different crystal forms, is a property of some molecules and molecular complexes. A single compound, like Delgocitinib, may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (*e.g.* measured by thermogravimetric analysis – "TGA", or differential scanning calorimetry – "DSC"), X-ray powder diffraction (XRPD) pattern, infrared absorption fingerprint, Raman absorption fingerprint, and solid state (^{13}C-) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

[0005] Different salts and solid state forms (including solvated forms) of an active pharmaceutical ingredient may possess different properties. Such variations in the properties of different salts and solid state forms may provide a basis for improving formulation, for example, by facilitating better processing or handling characteristics, improving the dissolution profile, or improving stability (polymorph as well as chemical stability) and shelf-life. These variations in the properties of different salts and solid state forms may also provide improvements to the final dosage form, for instance, if they serve to improve bioavailability. Different salts and solid state forms of an active pharmaceutical ingredient may also give rise to a variety of polymorphs or crystalline forms, which may in turn provide additional opportunities to use variations in the properties and characteristics of a solid active pharmaceutical ingredient for providing an improved product.

[0006] Discovering new salts and solid state forms of a pharmaceutical product can provide materials having desirable processing properties, such as ease of handling, ease of processing, storage stability, and ease of purification or as desirable intermediate crystal forms that facilitate conversion to other salts or polymorphic forms. New polymorphic forms and new salts of a pharmaceutically useful compound can also provide an opportunity to improve the performance characteristics of a pharmaceutical product (dissolution profile, bioavailability, etc.). It enlarges the repertoire of materials that a formulation scientist has available for formulation optimization, for example by providing a product with different properties, *e.g.*, a different crystal habit, higher crystallinity or polymorphic stability which may offer better processing or handling characteristics, improved dissolution profile, or improved shelf-life. For at least these reasons, there is a need for additional salts and solid state forms (including solvated forms) of Delgocitinib.

SUMMARY OF THE INVENTION

[0007] The present disclosure relates to Delgocitinib solid state forms, crystalline polymorphs thereof or amorphous form, to processes for preparation thereof, and to pharmaceutical compositions comprising solid state form thereof.

[0008] In particular, the present disclosure provides crystalline forms of Delgocitinib designated as Forms DL1 – DL13 and crystalline Delgocitinib hydrochloride (defined herein).

[0009] The present disclosure further provides process for preparing Delgocitinib and solid state forms or crystalline polymorphs thereof.

[0010] In another aspect, the present disclosure encompasses the above described solid state forms or crystalline polymorphs of Delgocitinib and crystalline Delgocitinib hydrochloride for use in the preparation of pharmaceutical compositions and/or formulations, preferably for use in medicine, preferably for treating atopic dermatitis (AD).

[0011] In another aspect, the present disclosure encompasses the use of any one of the above described solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride for the preparation of pharmaceutical compositions and/or formulations, preferably for use in medicine, preferably for treating atopic dermatitis (AD). In yet another embodiment, the present disclosure encompasses pharmaceutical compositions comprising any one of the solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride.

[0012] In a specific embodiment, the present disclosure encompasses pharmaceutical formulations comprising any one of the above described solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride, and at least one pharmaceutically acceptable excipient.

[0013] The present disclosure further encompasses processes to prepare said pharmaceutical formulations of Delgocitinib and Delgocitinib hydrochloride, comprising combining any one of the above described solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride, or pharmaceutical compositions comprising it, and at least one pharmaceutically acceptable excipient.

[0014] The solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride as defined herein as well as the pharmaceutical compositions or formulations comprising it can be used as medicaments, particularly for treating complement mediated diseases, preferably for the treatment of atopic dermatitis, comprising administering a therapeutically effective amount of the solid state form of the present disclosure, or at least one of the above pharmaceutical compositions or formulations, to a subject suffering from atopic dermatitis, or otherwise in need of the treatment.

[0015] The present disclosure also provides methods of treating atopic dermatitis, comprising administering a therapeutically effective amount of any one of the solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride of the present disclosure, or at least one of the above pharmaceutical compositions or formulations, to a subject suffering from atopic dermatitis, or otherwise in need of the treatment.

[0016] The present disclosure also provides the uses of the solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride of the present disclosure, or at least one of the above pharmaceutical compositions or formulations, for the manufacture of medicaments for treating atopic dermatitis.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Figure 1 shows an X-ray powder diffraction (XRPD) pattern for crystalline Delgocitinib Form DL1.

[0018] Figure 2 shows an XRPD pattern of crystalline Delgocitinib Form DL2.

[0019] Figure 3 shows an XRPD pattern of crystalline Delgocitinib Form DL3.

[0020] Figure 4 shows an XRPD pattern of crystalline Delgocitinib Form DL4.

[0021] Figure 5 shows an XRPD pattern of Amorphous Delgocitinib.

[0022] Figure 6 shows an XRPD pattern of crystalline Delgocitinib Form DL5.

[0023] Figure 7 shows an XRPD pattern of crystalline Delgocitinib Form DL6.

[0024] Figure 8 shows an XRPD pattern of crystalline Delgocitinib Form DL7.

[0025] Figure 9 shows an XRPD pattern of crystalline Delgocitinib Form DL8.

[0026] Figure 10 shows an XRPD pattern of crystalline Delgocitinib Form DL9.

[0027] Figure 11 shows an XRPD pattern of crystalline Delgocitinib Form DL10.

[0028] Figure 12 shows an XRPD pattern of crystalline Delgocitinib Form DL11.

[0029] Figure 13 shows an XRPD pattern of crystalline Delgocitinib Form DL12.

[0030] Figure 14 shows an XRPD pattern of crystalline Delgocitinib Form DL13.

[0031] Figure 15 shows an XRPD pattern of crystalline Delgocitinib hydrochloride.

[0032] Figure 16 shows an XRPD pattern of crystalline Delgocitinib Form Alpha.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0033] The present disclosure relates to solid state forms or crystalline polymorphs of Delgocitinib and Delgocitinib hydrochloride, to processes for preparation thereof and to pharmaceutical compositions comprising at least one of, or combination of these solid state forms. In particular, the present disclosure relates to solid state forms of Delgocitinib designated as Form DL1 - Form DL13 and Delgocitinib hydrochloride (defined herein).

[0034] The Delgocitinib solid state forms, according to the present disclosure may have advantageous properties selected from at least one of: chemical or polymorphic purity, flowability, solubility, dissolution rate, bioavailability, morphology or crystal habit, stability – such as chemical stability as well as thermal and mechanical stability with respect to polymorphic conversion, stability towards dehydration and/or storage stability, a lower degree of hygroscopicity, low content of residual solvents, adhesive tendencies and advantageous processing and handling characteristics such as compressibility, and bulk density.

[0035] A crystal form may be referred to herein as being characterized by graphical data "as depicted in" a Figure. Such data include, for example, powder X-ray diffractograms and solid state NMR spectra. As is well-known in the art, the graphical data potentially provides additional technical information to further define the respective solid state form (a so-called "fingerprint") which can not necessarily be described by reference to numerical values or peak positions alone. In any event, the skilled person will understand that such graphical representations of data may be subject to small variations, *e.g.*, in peak relative intensities and peak positions due to factors such as variations in instrument response and variations in sample

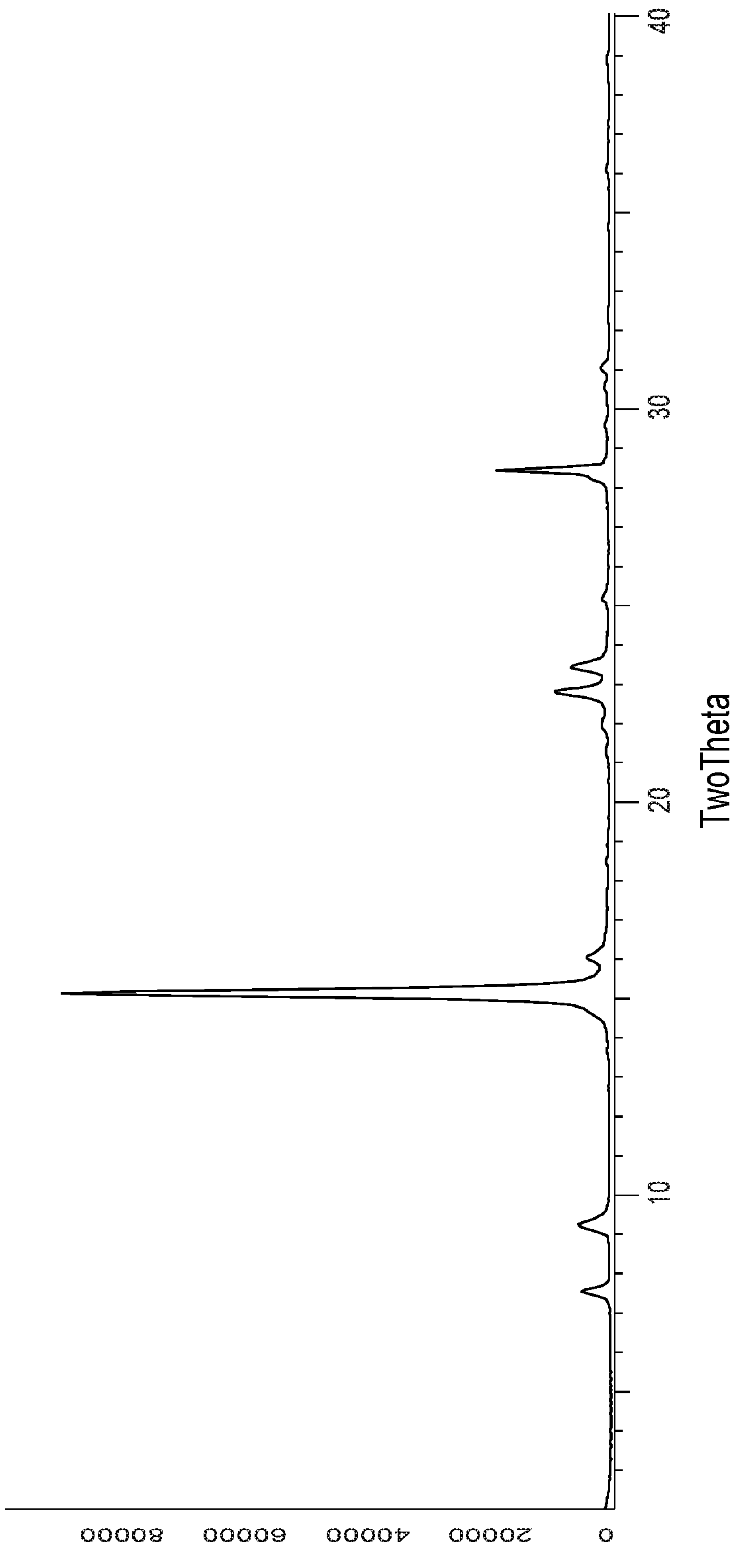


Figure 4. An XRPD pattern for crystalline Delgocitinib Form DL4.

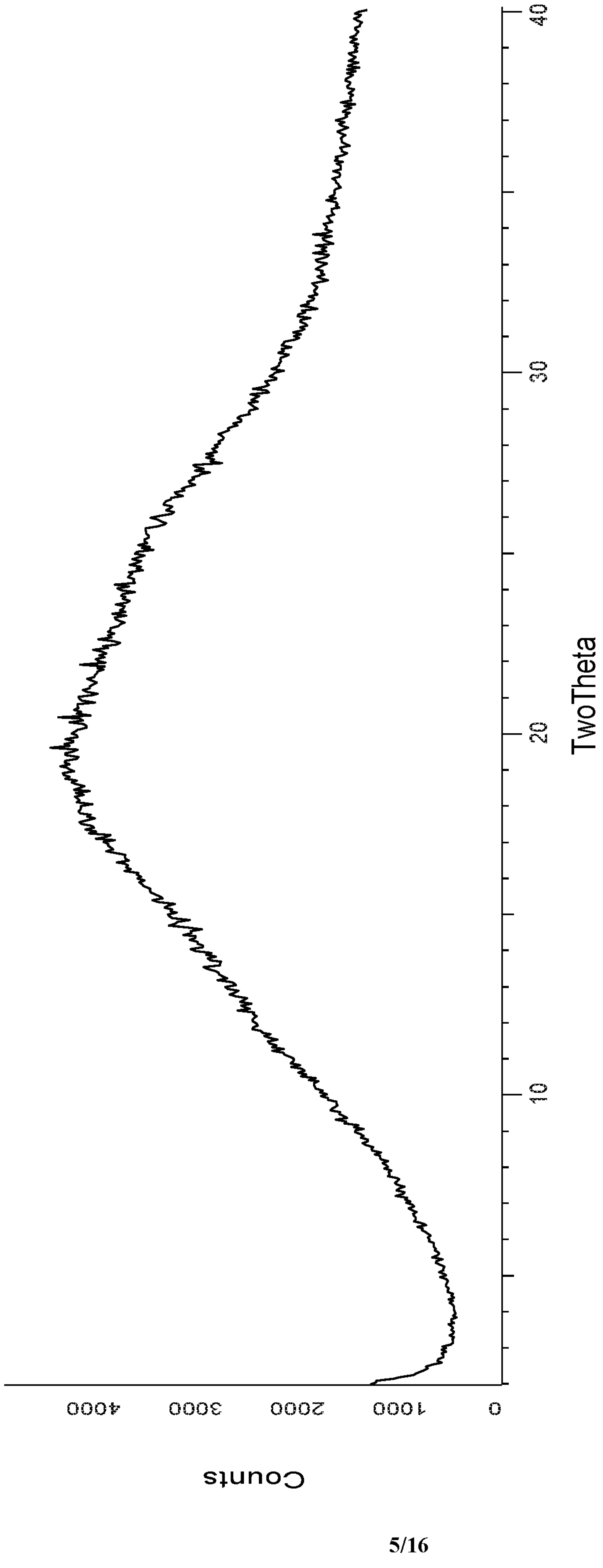


Figure 5. An XRPD pattern for amorphous Delgocitinib.

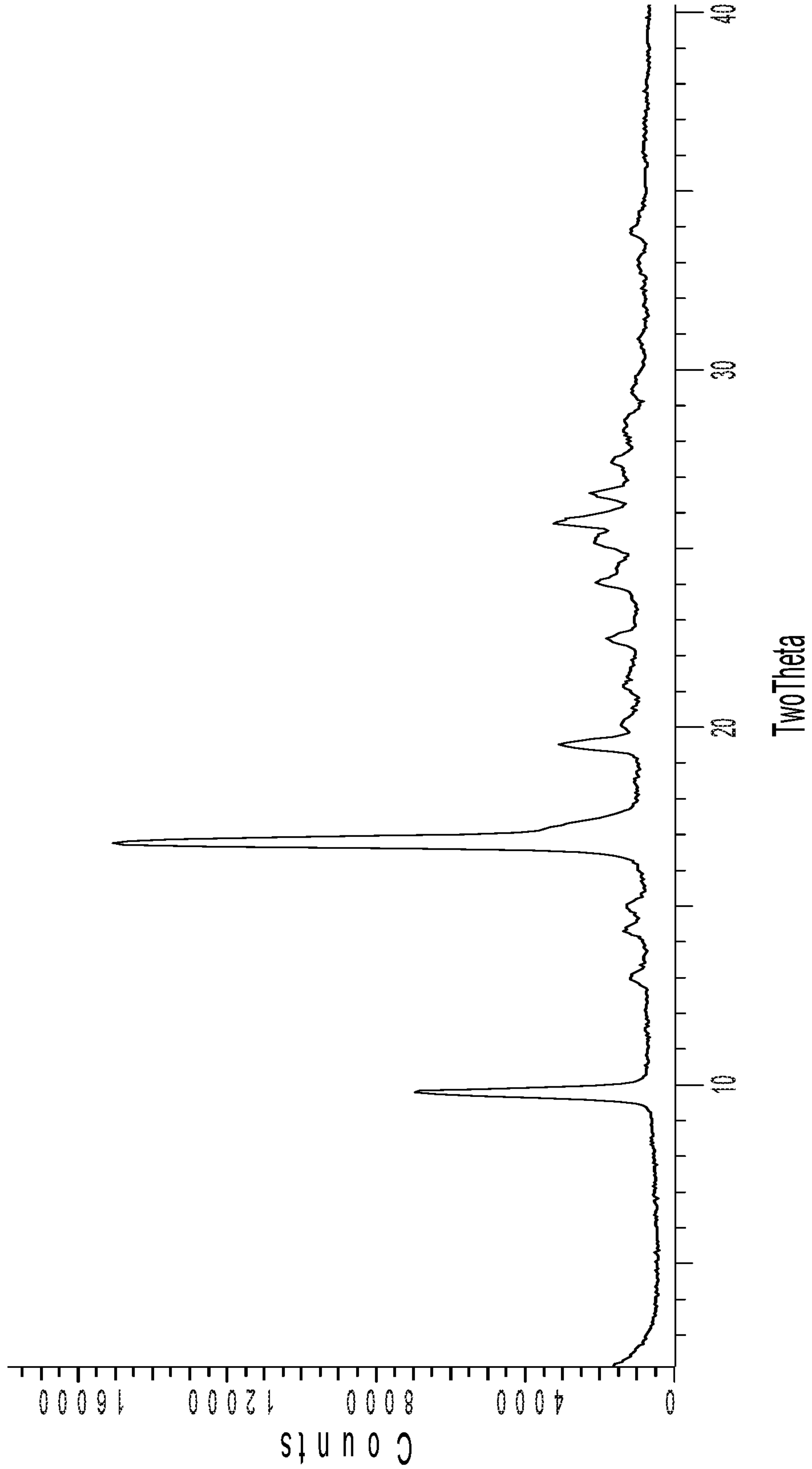


Figure 6. An XRPD pattern for crystalline Delgocitinib Form DL5.

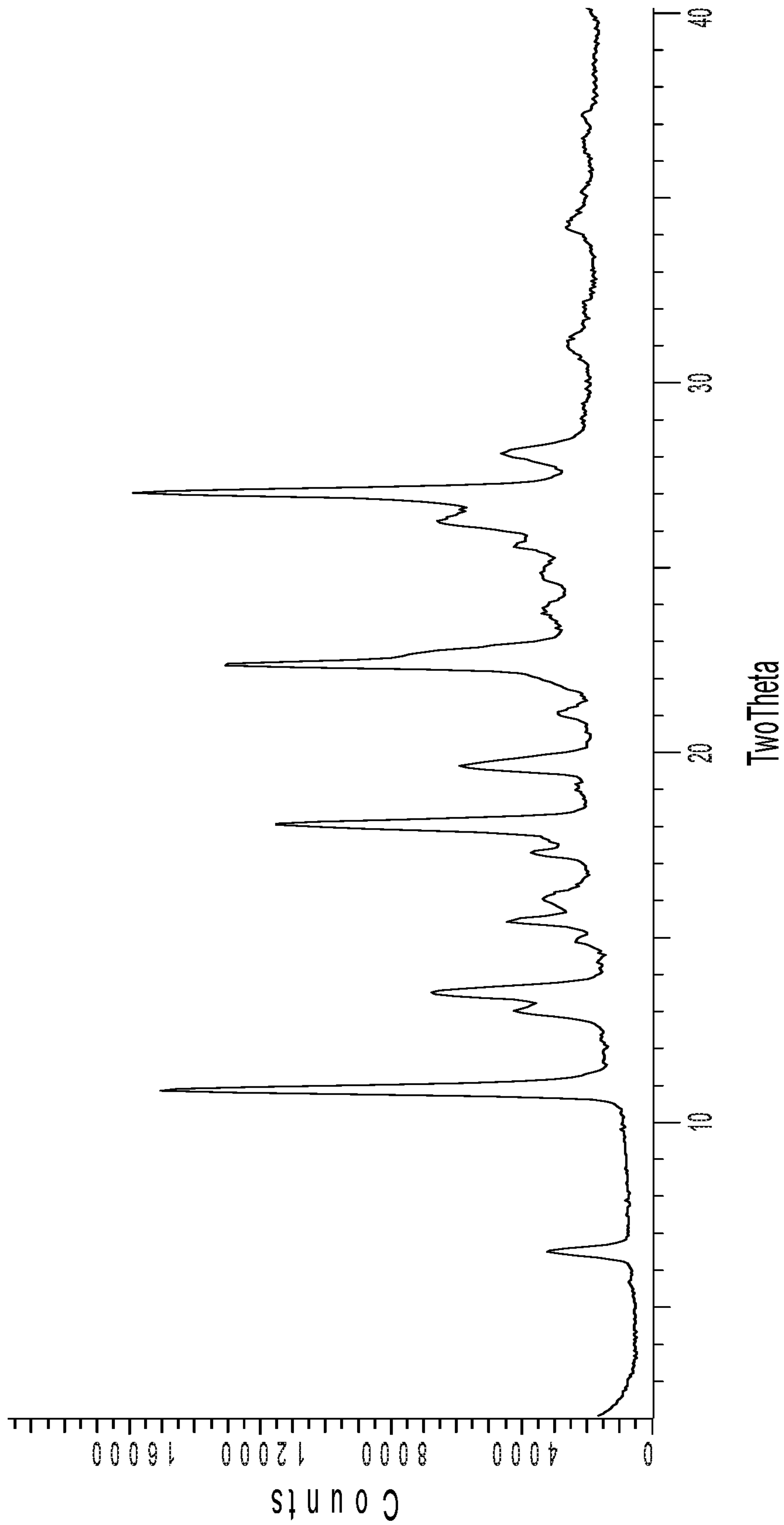


Figure 7. An XRPD pattern for crystalline Delgocitinib Form DL6.

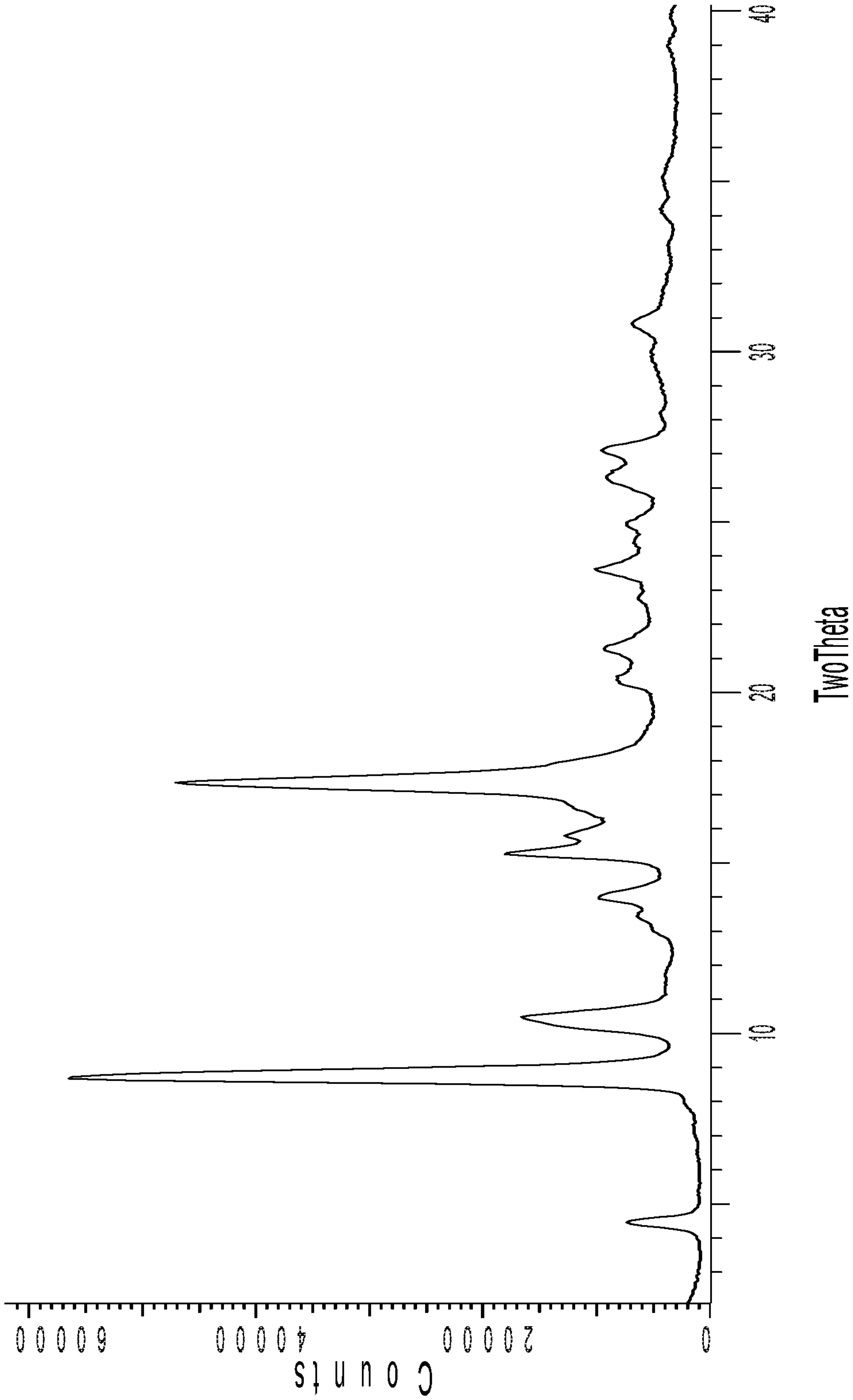


Figure 8. An XRPD pattern for crystalline Delgocitinib Form DL7.

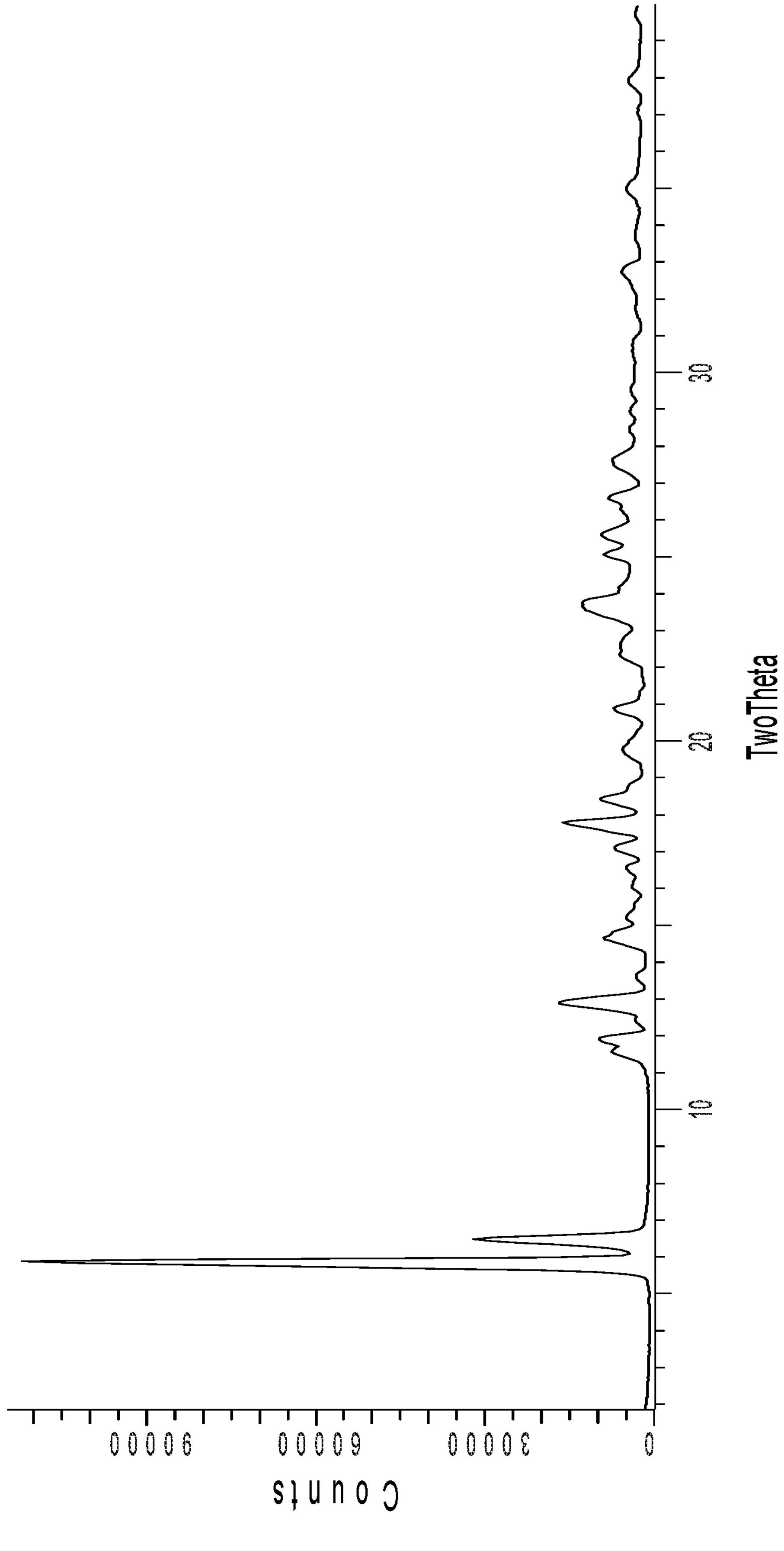


Figure 9. An XRPD pattern for crystalline Delgocitinib Form DL8.

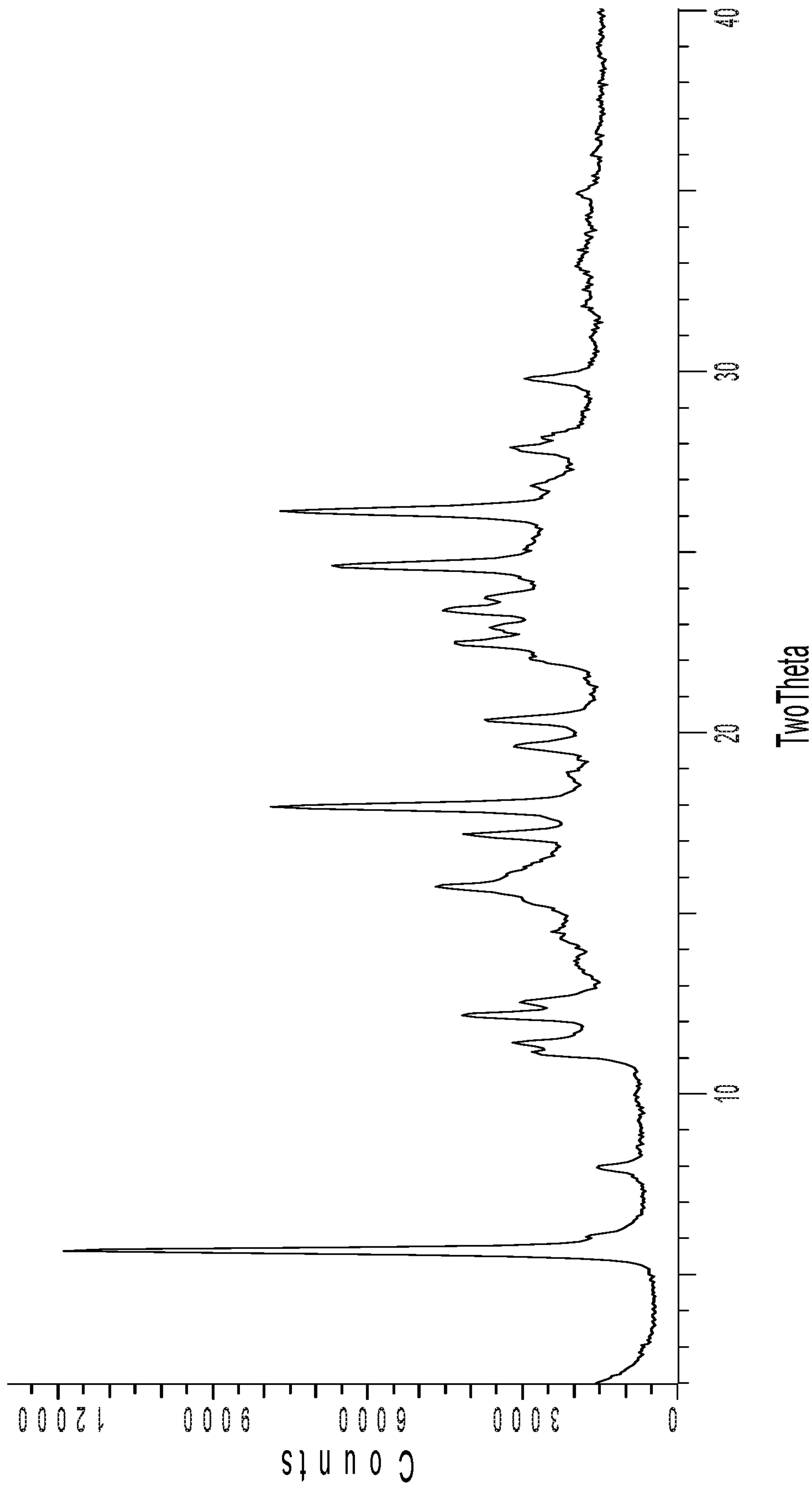


Figure 10. An XRPD pattern for crystalline Delgocitinib Form DL9.

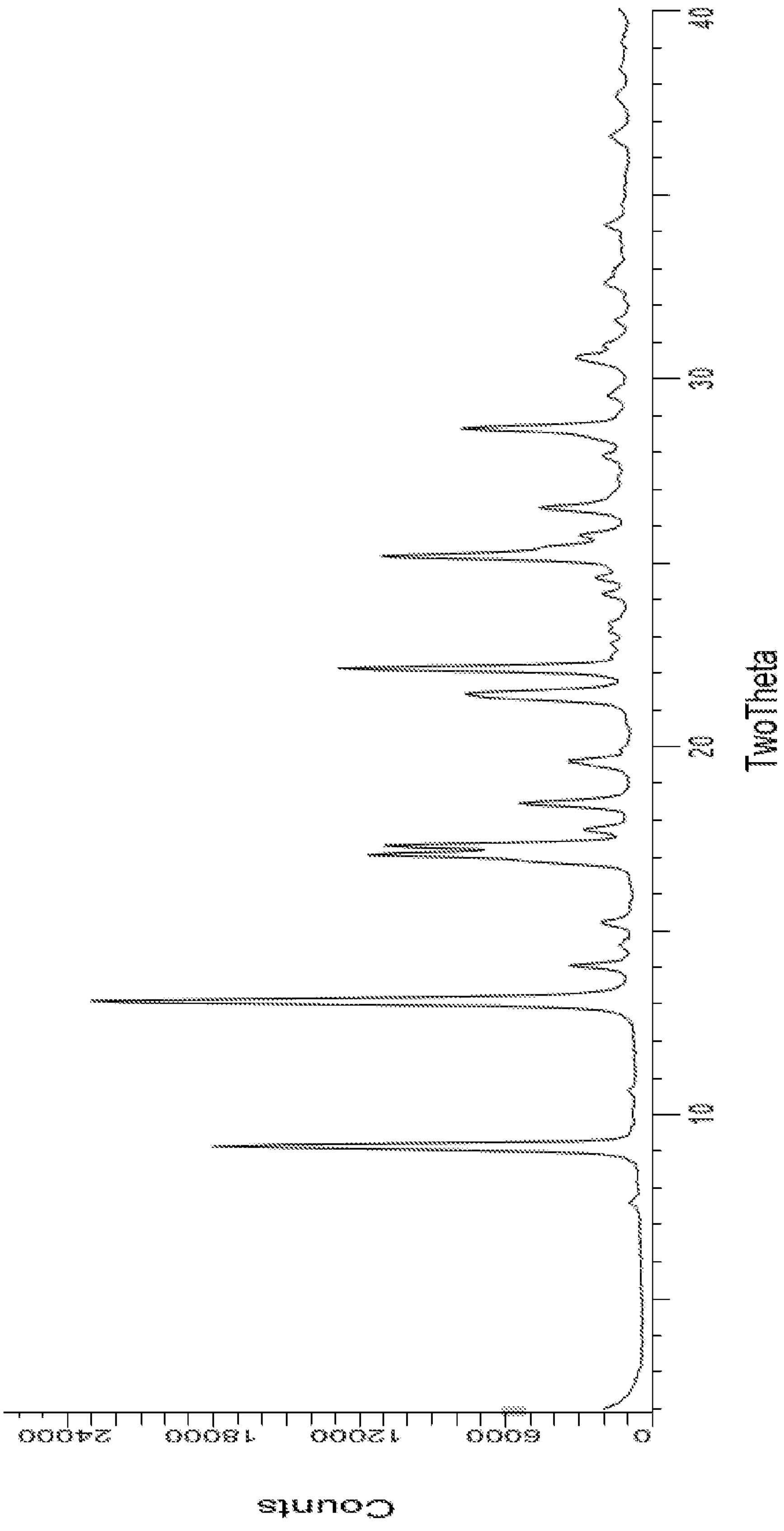


Figure 11. An XRPD pattern of crystalline Delgocitinib Form DL10.

INTERNATIONAL SEARCH REPORT

International application No
PCT/IB2024/051909

A. CLASSIFICATION OF SUBJECT MATTER
INV. A61P17/06 C07D487/10 A61K31/519
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
A61P C07D

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, CHEM ABS Data, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2018/117153 A1 (JAPAN TOBACCO INC [JP]) 28 June 2018 (2018-06-28) the whole document -----	1-4, 21-31

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
15 May 2024	30/07/2024

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Koch, Kristian

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2024/051909

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:
1-4(completely); 21-31(partially)

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-4(completely); 21-31(partially)

   Crystalline Form DL2 of Delgocitinib.

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2. claims: 5-8(completely); 21-31(partially)

   Crystalline Form DL1 of Delgocitinib.

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3. claims: 9-12(completely); 21-31(partially)

   Crystalline Form DL3 of Delgocitinib.

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4. claims: 13-16(completely); 21-31(partially)

   Crystalline Form DL10 of Delgocitinib.

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5. claims: 17-20(completely); 21-31(partially)

   Crystalline Form DL11 of Delgocitinib.

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/IB2024/051909

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2018117153 A1	28-06-2018	AU 2017380213 A1	23-05-2019
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